Non-Markovian Structure as a Forced Memory-Kernel Layer of Event-Density Theory

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Abstract

We extend the Event-Density (ED) program from its Phase-2 form-level closure to a kernel-level memory-structure layer. Phase-2 (Arc R relativistic kinematics, Arc M chain-mass, Arc Q QFT extension) closed the structural framework for ED's quantum sector under implicit Markovian assumptions; Arc N tests whether those assumptions are primitive-derived or merely tractability-driven. We address the central question (Q-N): *does ED admit or require non-Markovian structure in chain evolution or multi-chain interactions?* Through a five-stage analysis (N.0 scoping, N.1 catalogue, N.2 FORCED evaluation, N.3 REFUTED evaluation, N.4 cross-arc implications, N.5 synthesis), we establish three principal results. **Theorem N1 (Finite-Width Vacuum Memory Kernel forced):** the vacuum-response kernel $K_{\text{vac}}(x - x')$ is structurally forced at primitive level to have finite, non-zero, decaying, Lorentz-scalar, sub-power-law-2 form, by the joint action of UV-FIN (Arc Q Theorem Q3), Primitive 01 event-discreteness, and Primitive 13 finite proper-time intervals. Three cascading FORCED-conditional items (vacuum-induced bandwidth memory N1-E, vacuum-modulated commitment memory N2-E, vacuum-mediated adjacency memory N3-D) plus V5 cross-chain-correlation existence inherit forced status via the V1 vacuum-coupling channel. Seven REFUTED determinations under the C1 (Lorentz covariance), C2 (spin-statistics), C3 (UV-FIN) constraint framework establish that admissible non-Markovianity is bounded, not arbitrary. Eleven catalogue items plus four V-sector specific-form sub-cases remain ADMISSIBLE-NOT-FORCED. No Phase-2 verdict is overturned; three Phase-2 closures (Arc M σ_τ, Arc Q.5 vacuum polarisation, Arc Q.8 cosmological-Λ) are structurally enriched; one new CANDIDATE (vacuum-mediated additional CP phases beyond the Jarlskog count) is opened. Phase-3 (ED $\to$ GR coupling) is substantively unblocked with three new structural input channels. Theorem N1 takes its place as ED's eighth FORCED structural theorem and first memory-kernel-level structural prediction, completing the quantum-sector picture at the kernel level alongside Phase-2's form-level theorems while preserving the *form-FORCED, value-INHERITED* methodology established across Arcs R, M, and Q.

1 Introduction

The Event-Density (ED) program builds quantum and gravitational physics as forced structural consequences of a small primitive stack on a 3+1-dimensional event manifold. Phase-1 [1] derived non-relativistic single-particle quantum mechanics from the participation measure $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$, the four-band bandwidth decomposition (Primitive 04), and commitment dynamics (Primitives 10–11). Phase-2 extended this foundation through three connected arcs: Arc R [2] derived Klein–Gordon, the spin–statistics theorem, the Cl(3,1) Clifford-algebra frame, and the Dirac equation; Arc M [3] established that ED forces the *form* of mass structure but inherits all numerical mass values, ratios, and hierarchies; Arc Q [4] extended ED to the QFT regime, producing GRH (gauge-field-as-rule-type) unconditional FORCED, canonical (anti-)commutation FORCED, and UV-FIN (primitive-level UV finiteness) FORCED.

Phase-2's three arcs closed under implicit *Markovian* assumptions. Stage R.1's minimal coupling, Arc M's σ_τ master formula, Arc Q's canonical equal-time commutation relations,

and Arc Q.8's effective vacuum response were each derived assuming locality in time — the assumptions were never primitive-derived; they were tractability-driven choices.

Arc N opens the question of whether those assumptions are required at primitive level. The central question is:

> **(Q-N)** Does ED admit or require non-Markovian structure in chain evolution or multi-chain interactions?

This paper presents Arc N's five-stage closure: N.0 scoping [5], N.1 catalogue [6], N.2 FORCED evaluation [7], N.3 REFUTED evaluation [8], N.4 cross-arc implications [9], and N.5 synthesis [10] (the authoritative source for the present manuscript).

The headline verdict, in plain language: **ED is Markov-compatible but not Markov-forcing.** Most non-Markovian extensions are admissible structurally but not required. One memory-kernel-level structure is, however, structurally FORCED at primitive level — the finite-width vacuum-response kernel V1, established as Theorem N1. The forcing argument is downstream of UV-FIN: a δ-function vacuum response would amplify high-frequency contributions and threaten primitive-level UV finiteness; primitive event-discreteness imposes a natural cutoff. Theorem N1 specifies the structural form of the resulting finite-width kernel.

Beyond Theorem N1, four cascading items inherit FORCED-conditional status via the V1 vacuum-coupling channel: vacuum-induced bandwidth memory (N1-E), vacuum-modulated commitment memory (N2-E), vacuum-mediated adjacency memory (N3-D), and V5 cross-chain correlation existence. Seven REFUTED determinations under the C1/C2/C3 constraint framework eliminate specific non-Markovian forms — confirming that admissible non-Markovianity is bounded.

Phase-2 verdicts are unaffected. Three Phase-2 closures are structurally *enriched* with kernel-level detail, but no verdict is overturned. One new CANDIDATE — vacuum-mediated additional CP phases beyond the Jarlskog count — is opened in Arc Q.6's territory. Phase-3 (ED $\to$ GR coupling) is substantively unblocked with three new structural input channels arising from Theorem N1's curved-spacetime extension.

The paper is organised as follows. Section 2 reviews ED primitives and Phase-2 structural theorems relevant to Arc N. Section 3 presents the Arc N roadmap and methodology. Section 4 summarises the catalogue of admissible non-Markovian structures. Section 5 states and proves Theorem N1. Section 6 presents REFUTED and ADMISSIBLE structures. Section 7 evaluates cross-arc implications. Section 8 presents the Phase-3 hand-off. Section 9 discusses methodological consistency. Section 10 concludes.

2 Background and Position in the ED Program

2.1 ED primitives relevant to Arc N

- **Primitive 01 (micro-event):** discrete events on the 3+1-dimensional event manifold; supplies the event-discreteness scale ℓ_{ED}.
- **Primitive 04 (bandwidth):** four-band decomposition $b_K = b_K^{\mathrm{int}} + b_K^{\mathrm{adj}} + b_K^{\mathrm{env}} + b_K^{\mathrm{com}}$ with bounded amplitudes per primitive event.
- **Primitive 06 (four-gradient + Lorentz covariance):** supplies the C1 constraint.
- **Primitive 10 (individuation):** threshold separating distinct chains; supports Pauli exclusion via vanishing-on-coincidence for Case-R rule-types.
- **Primitive 11 (commitment):** discrete events along γ_K. Substrate for the history-bearing process structure.

- **Primitive 13 (relational timing):** finite proper-time intervals τ_K along γ_K. Supplies the second leg of the UV-finiteness argument.

A primitive-by-primitive review confirms ED's general status: **primitives 01–13 admit non-Markovian structure as a consistent extension but do not generally force it.** The exception, discussed in Section 5, is the joint action of Primitives 01 + 04 + 13 under the UV-FIN constraint, which together force vacuum-response kernels to have finite width.

2.2 Phase-2 forced theorems

Arc N inherits seven FORCED structural theorems from Phase-1 + Phase-2:

#	Theorem	Source / Type
1	Spin–statistics $\eta = (-1)^{2s}$	R.2.5 / Theorem-level
2	Cl(3,1) frame uniqueness	R.2.4 / Algebra-level
3	Anyon prohibition in 3+1D	R.2.3 / Topology-level
4	Dirac equation emergence	R.3 / Dynamical-equation-level
5	GRH unconditional FORCED	Q.1 + Q.2/3/7/8 / Existence-level
6	Canonical (anti-)commutation FORCED	Q.7 / Operator-level
7	UV-FIN FORCED	Q.8 / Bound-level

Each operates at the **form-level** of ED's quantum sector. None operates at the kernel level — the specific mechanism by which UV-FIN is realised in vacuum response.

2.3 Arc N as kernel-level complement to Phase-2

Arc N positions itself as a **kernel-level** complement to Phase-2's form-level theorems. Phase-2 says "primitive-level multi-chain participation integrals are finite"; Arc N says "the kernel mechanism by which finiteness is realised is V1 with finite, non-zero, decaying, sub-power-law-2 form."

This structural division mirrors the Arc M precedent — Arc M established the form of the σ_τ mass structure but inherited specific values; Arc N establishes the form of the vacuum-response kernel but inherits specific functional realisations within that form. The methodological framing — *form-FORCED, value-INHERITED* — extends naturally from Arc M to Arc Q to Arc N.

3 Arc N Roadmap and Methodology

Arc N proceeds through five substages:

- **N.0 (scoping)** [5]: frames (Q-N), enumerates three candidate mechanisms by sector (N1 bandwidth-memory, N2 commitment-memory, N3 adjacency-memory), three potential primitive sources (S1–S3), three potential constraints (C1–C3). Honest expected-verdict prior recorded.

- **N.1 (catalogue)** [6]: enumerates 20 admissible non-Markovian structures across N1, N2, N3, V sectors. No FORCED/REFUTED evaluation — intentional pre-judgment enumeration.

- **N.2 (forced evaluation)** [7]: V1 unconditionally FORCED; N1-E, N2-E, N3-D FORCED-conditional-on-V1; V5 existence FORCED-conditional with amplitudes inherited.

- **N.3 (refuted evaluation)** [8]: 7 REFUTED determinations under C1/C2/C3.

- **N.4 (cross-arc implications)** [9]: no Phase-2 verdict overturned; three closures structurally enriched; one new CANDIDATE C-N4.1 opened.

- **N.5 (synthesis)** [10]: integration of N.0–N.4 into final Arc N verdict.

Methodological discipline preserved across all five stages: clear separation of FORCED / CANDIDATE / REFUTED / ADMISSIBLE / INHERITED; honest prior recorded; *form-FORCED, value-INHERITED* framing applied at the kernel level; cross-arc implications explicitly evaluated.

4 Catalogue of Non-Markovian Structures

Stage N.1 [6] enumerated 20 admissible non-Markovian structures across four sectors. We summarise without re-deriving.

4.1 Sector N1 — bandwidth-memory (5 items)

- **N1-A** Finite-width memory kernels in b^{env}.
- **N1-B** History-weighted spectral-rate terms in σ_τ.
- **N1-C** Multi-scale bandwidth memory.
- **N1-D** Cross-band memory coupling, including potential Case-P↔Case-R cross-band coupling.
- **N1-E** Vacuum-induced bandwidth memory inheriting kernel structure from V1.

4.2 Sector N2 — commitment-memory (5 items)

- **N2-A** Thresholds depending on past event density.
- **N2-B** Hysteresis-style commitment with delayed individuation.
- **N2-C** Memory-weighted commitment spacing.
- **N2-D** Multi-chain commitment correlations.
- **N2-E** Vacuum-modulated commitment memory.

4.3 Sector N3 — adjacency-memory (5 items)

- **N3-A** Time-lagged adjacency graphs.
- **N3-B** History-dependent adjacency weights.
- **N3-C** Multi-chain (3+) correlation kernels.
- **N3-D** Vacuum-mediated adjacency memory.
- **N3-E** Non-local adjacency propagation (catalogued for completeness; expected REFUTED).

4.4 Sector V — vacuum-response (5 items)

- **V1** Finite-width vacuum memory kernel (general functional form).
- **V2** Exponential-decay vacuum response.
- **V3** Power-law vacuum response (admissible only below critical exponent).
- **V4** Multi-scale vacuum memory.
- **V5** Vacuum-induced cross-chain correlations.

5 Theorem N1: Finite-Width Vacuum Memory Kernel forced

5.1 Statement

Theorem 5.1 (Finite-Width Vacuum Memory Kernel FORCED, N1)**.** *The vacuum-response kernel $K_{\text{vac}}(x-x')$ acting on chain participation perturbations is structurally* FORCED *at primitive level to satisfy:*

(i) ***Finite width:*** *$K_{\text{vac}}(x-x')$ is non-singular at $x = x'$ (rules out δ^4-function form).*

(ii) ***Decaying support:*** *$K_{\text{vac}}(x-x') \to 0$ as $|x-x'| \to \infty$ (rules out constant kernels).*

(iii) ***Lorentz-scalar form:*** *K_{vac} depends only on Lorentz invariants of $x - x'$.*

(iv) ***Bounded short-time behaviour:*** *kernel growth is sub-power-law-2 in the proper-time-invariant separation in 3+1D (rules out $\alpha \geq 2$ power-law forms).*

Specific functional forms — exponential decay (V2), restricted power-law $\alpha < 2$ (V3), multi-scale composition (V4) — are admissible specific realisations within the FORCED *class. Specific values (kernel widths, decay constants, multi-scale weights) are* INHERITED.

5.2 Proof sketch

The proof proceeds through three structural-argument legs.

Leg 1 — UV-FIN forces non-zero finite width. Suppose the kernel were a δ-function: $K_{\text{vac}}(x-x') = c_0\delta^4(x-x')$. In Fourier space, $\widetilde{K_{\text{vac}}}(\omega, \mathbf{k}) = c_0$ — constant across all frequencies and wavenumbers. In any continuum-approximation loop integral involving the vacuum response, this constant frequency-weighting amplifies high-frequency contributions, threatening UV-FIN (Arc Q Theorem Q3 [4]).

Primitive 01 event-discreteness imposes a primitive-level event-resolution scale ℓ_{ED} below which the manifold structure is not continuum-like. Primitive 13 supplies finite proper-time intervals between commitment events. Jointly, vacuum response cannot resolve perturbations at sub-primitive resolution; the kernel must be smoothed at the ℓ_{ED} scale.

The δ-limit is therefore REFUTED by C3. The finite-width condition (i) follows.

Leg 2 — Lorentz covariance + locality forces decaying support. Suppose the kernel were a constant: $K_{\text{vac}}(x-x') = c_\infty$ for all x, x'. While trivially Lorentz-scalar, this implies infinite correlation length: every event is correlated with every other event with equal weight, including arbitrary spacelike separations.

Such a kernel is structurally pathological. It does not respect causal-structure constraints embedded in Lorentz covariance, and is incompatible with Primitive 11 commitment-event locality. Any structurally consistent finite-width kernel must have *some* decay with separation.

The infinite-width limit is therefore REFUTED by C1 (in its locality-respecting interpretation). The decaying-support condition (ii) follows.

Leg 3 — UV-FIN refinement forces sub-power-law-2 decay. Power-law kernels $K_{\text{vac}} \propto |x-x'|^{-2\alpha}$ in 3+1D produce UV-divergent integrals when $2\alpha \geq 4$, i.e., $\alpha \geq 2$. The integral

$$\int d^4z\, |z|^{-2\alpha} \cdot \phi(z), \tag{1}$$

with ϕ a bounded test function, converges at small $|z|$ iff $\alpha < 2$ in 3+1D. Power-law kernels with $\alpha \geq 2$ are REFUTED by C3.

The sub-power-law-2 condition (iv) follows.

Synthesis. The three legs jointly force K_{vac} to satisfy conditions (i)–(iv). The Lorentz-scalar form (iii) follows from C1 directly. Specific functional forms within the bounded class are admissible specific realisations; specific values INHERITED. □

5.3 Dependencies

Theorem N1 depends on:

- Primitive 01 (event-discreteness): supplies natural-cutoff scale ℓ_{ED}.
- Primitive 04 (bounded bandwidth): bounds amplitude content for finiteness.
- Primitive 06 (Lorentz covariance): supplies C1 constraint.
- Primitive 11 (commitment-event locality): supplies C1-locality refinement.
- Primitive 13 (finite proper-time intervals): supplies second leg of finiteness.
- Arc Q Theorem Q3 (UV-FIN): primitive-level UV finiteness driving V1's necessity.
- Arc Q Stage Q.8 (effective-vacuum framework): supplies the structural object on which V1 acts.

V1's forcing is downstream of UV-FIN. UV-FIN is the more general theorem; V1 is its specific realisation at the vacuum-response level.

5.4 Significance

Theorem N1 is **ED's first memory-kernel-level structural theorem**. It takes its place as the eighth FORCED structural theorem in the ED inventory, complementing the seven Phase-2 form-level theorems with kernel-level content.

5.5 Cascading forced-conditional items

Theorem N1 propagates FORCED-conditional status to four secondary catalogue items:

- **N1-E (vacuum-induced bandwidth memory):** b^{env} band inherits memory-kernel structure from V1.
- **N2-E (vacuum-modulated commitment memory):** commitment dynamics interact with the local vacuum state.
- **N3-D (vacuum-mediated adjacency memory):** same-vacuum-sector chains acquire correlated b^{env} content.
- **V5 (vacuum-induced cross-chain correlations):** existence FORCED-conditional on V1; amplitudes INHERITED.

Since V1 is unconditionally FORCED, these conditional statuses are effectively unconditional; the conceptual hierarchy is retained.

6 refuted and admissible Structures

6.1 refuted determinations

Stage N.3 [8] produced seven REFUTED determinations:

ID / Sub-case	Constraint	Brief reason
N1-D, P↔R	C2	Cross-class kernel produces η-mixing; $\pi_1 = \mathbb{Z}_2$ forbids intermediate
N2-B, Case-R	C2	Two Case-R chains coexist within delay window; violates Pauli
N3-E, frame-dep.	C1	Preferred-frame correlations forbidden by Lorentz covariance
V3, $\alpha \geq 2$	C3	$\int d^4z\|z\|^{-2\alpha}$ diverges in 3+1D
V1-δ	C3	δ-response amplifies UV
V1-∞	C1	Constant kernel breaks locality
V5 large-amp/long-range	C1, C3	Infinite correlation length / insufficient decay

The pattern: 3 C1 violations, 2 C2 violations, 3 C3 violations (with V5 spanning C1+C3). All three constraints are structurally active. The C1/C2/C3 framework is exhaustive.

6.2 V1 admissibility class is bounded both ways

V1-δ and V1-∞ refutations together establish that V1's admissibility class is **bounded**: kernels must have finite, non-zero, decaying width. The bounds are derived consequences of C1 + C3 — not additional postulates.

6.3 admissible-not-forced structures

Eleven catalogue items plus four V-sector specific-form sub-cases pass Stage N.3 unrefuted:

ID	Description	Influences
N1-A	Finite-width b^{env} kernels	Arc M, Q.5
N1-B	History-weighted σ_τ spectral-rate	Arc M
N1-C	Multi-scale bandwidth memory	Arc M, Phase-3
N1-D, same-class	Same-class cross-band coupling	Arc M
N2-A	Density-dependent thresholds	Arc Q.7
N2-B, Case-P	Hysteresis on Case-P	Arc Q.7
N2-C	Memory-weighted commitment spacing	Arc Q.5, Q.7
N2-D	Multi-chain commitment correlations	Arc Q.6, Q.7
N3-A	Time-lagged adjacency graphs	Arc Q.6, Q.7
N3-B	History-dep. adjacency weights	Arc Q.6
N3-C	Multi-chain (3+) kernels	Arc Q.6, Phase-3
V2	Exponential-decay vacuum response	Arc Q.5, Q.8, Phase-3
V3 ($\alpha < 2$)	Restricted power-law	Arc Q.5, Phase-3
V4	Multi-scale vacuum memory	Arc Q.5, Phase-3
V5 (bounded)	Bounded cross-chain correlations	Arc Q.6, Phase-3

6.4 Bounded non-Markovianity, not arbitrary

The combined Stage N.2 + N.3 result establishes that **admissible non-Markovianity in ED is bounded, not arbitrary.** Not every memory-kernel form is admissible. The admissibility class is well-defined and bounded; admissible non-Markovian structures inherit Phase-2 structural constraints automatically.

7 Cross-Arc Implications

Stage N.4 [9] evaluated the impact of FORCED V1 + cascades + ADMISSIBLE structures on Phase-2 closures. **No Phase-2 verdict is overturned by Arc N.**

7.1 Arc M (mass structure)

Arc M's σ_τ master formula [3],

$$\sigma_\tau = \hbar \cdot \sqrt{\sum_X w_\tau^X \cdot \langle (\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X) \rangle_\tau}, \tag{2}$$

acquires vacuum-coupled detail under N1-E. The form is preserved (SC1 Lorentz-scalar; SC2 amplitude-invariance). H1-dominant verdict — *ED fixes the structure of mass but not its values* — preserved.

7.2 Arc Q.5 (radiative corrections)

Vacuum polarisation transversality (FORCED by GRH-3 [4]) preserved under V1 modification. UV-FIN remains FORCED; V1 is its vacuum-response realisation. Tree-level $g = 2$ unaffected. Schwinger-sign CANDIDATE-plausible refined under V1 but not promoted.

7.3 Arc Q.6 (generations and mixing)

Generation count = 3 remains REFUTED-as-forced and EMPIRICAL [4]. Mixing-matrix existence and CP-phase existence (Jarlskog count) remain FORCED. The substantive enrichment:

(C-N4.1, vacuum-mediated additional CP phases): *V5 cross-chain correlations under Theorem N1 introduce complex-phase content into the effective Yukawa coupling matrix at vacuum-coupled scales; for $n \geq 3$ generations, this may force additional CP-violation structure beyond Jarlskog.* CANDIDATE; specific number depends on V1/V5 kernel form (INHERITED).

7.4 Arc Q.7 (second quantisation)

Canonical (anti-)commutation FORCED at equal-time coincident-point level (Theorem Q2 [4]) preserved. Pauli exclusion at every event preserved. Unequal-time commutators acquire memory-kernel content from V1:

$$\langle 0|T\{a_\tau(x)a_\tau^\dagger(x')\}|0\rangle = G_F^{(0)}(x-x') + \int K_{\text{vac}}(x-y) \cdot G_F^{(0)}(y-x')\, d^4y \cdot \mathcal{O}(\text{higher order}). \tag{3}$$

This is a structural extension of Arc Q.7 content.

7.5 Arc Q.8 (vacuum structure)

Multi-rule-type vacuum factorisation $|0\rangle_{\text{full}} = \bigotimes_\tau |0\rangle_\tau$ preserved. GRH unconditional FORCED unchanged (and reinforced). The cosmological-Λ divergence-form dissolution from Arc Q.8 gains explicit kernel-integral structure under Theorem N1:

$$\Lambda_{\text{primitive}} \sim \int K_{\text{vac}}(x-x') \cdot \rho_{\text{vac energy}}(x')\, d^4x'. \tag{4}$$

The "what numerical structure determines Λ?" question reframes to "what is the V1 kernel form, and what is the integration domain at cosmological scales?" Specific Λ value remains INHERITED.

8 Phase-3 Hand-Off

Arc N supplies three substantive structural-input channels for Phase-3 (ED $\rightarrow$ GR coupling).

8.1 Λ as V1-kernel integral

The cosmological-Λ value is identifiable as the integral of V1 vacuum-response kernel over cosmological-scale spacetime. Specific value INHERITED (depends on V1 kernel form + cosmological-boundary content).

8.2 Curvature–vacuum coupling candidate

In curved spacetime, V1 generalises via Synge's world function $\sigma(x, x')$ in place of $(x - x')^2$. The curvature-coupling structure may produce new structural terms in the effective ED-GR action — a Phase-3 evaluation deliverable.

8.3 Empirical signatures

- **Dispersion-relation modifications.** V1's finite-width kernel introduces frequency-dependent structure into vacuum response, producing small modifications to photon and graviton dispersion at extreme high frequencies. Possibly testable via cosmic-ray observations or extreme-precision photon timing.
- **Cosmological-correlation persistence.** V5 cross-chain correlations imply vacuum-mediated correlations between distant cosmological structures.

Both routes inherit V1/V5 kernel structure; specific values INHERITED.

9 Discussion

9.1 ED is Markov-compatible but not Markov-forcing

Arc N's primitive-level analysis confirms the opening claim: Primitives 01–13 admit non-Markovian structure as a consistent extension but do not generally force it. Phase-2's implicit Markovian assumptions were tractability-driven choices, not primitive-derived consequences.

The exception is V1: when UV-FIN is taken as a primitive-level constraint, the joint action of UV-FIN + Primitive 01 + Primitive 13 forces vacuum-response kernels to have finite width.

9.2 Form-forced, value-inherited preserved

Arc N preserves the methodological framing established in Arc M [3] and Arc Q [4]:

- **Form-forced:** V1 kernel class is structurally required; cascading mechanisms structurally required; admissibility bounds structurally derived.
- **Value-inherited:** specific functional form, timescales, amplitudes, Λ magnitude all inherited.

9.3 Arc N completes the quantum-sector picture at the kernel level

Pre-Arc-N, the ED quantum-sector picture was structurally complete at the form level but kernel-level unspecified. Arc N closes this gap: V1 is the kernel mechanism. Together with cascading items, V1 supplies the kernel-level structural detail that makes UV-FIN concrete.

9.4 Integration into the ED structural inventory

Theorem N1 is the eighth FORCED structural theorem in ED's inventory:

#	Theorem	Type
1	Spin–statistics $\eta = (-1)^{2s}$	Theorem-level
2	Cl(3,1) frame uniqueness	Algebra-level
3	Anyon prohibition in 3+1D	Topology-level
4	Dirac equation emergence	Dynamical-equation-level
5	GRH unconditional FORCED	Existence-level
6	Canonical (anti-)commutation FORCED	Operator-level
7	UV-FIN FORCED	Bound-level
8	**V1 finite-width vacuum kernel forced**	**Memory-kernel-level**

The five-arc structure — Phase-1 + Arc R + Arc M + Arc Q + Arc N — is **complete at primitive level** for ED's quantum sector. Phase-3 (ED → GR coupling) opens with this complete foundation.

10 Conclusion

Arc N closes with one FORCED structural theorem — Theorem N1 (Finite-Width Vacuum Memory Kernel FORCED) — established at primitive level by the joint action of UV-FIN, Primitive 01 event-discreteness, and Primitive 13 finite proper-time intervals. The V1 admissibility class is bounded both ways by V1-δ and V1-∞ refutations to finite, non-zero, decaying, Lorentz-scalar, sub-power-law-2 form. Four cascading FORCED-conditional items inherit forced status via the V1 vacuum-coupling channel.

Seven REFUTED determinations under C1/C2/C3 establish that admissible non-Markovianity is bounded, not arbitrary. Eleven catalogue items plus four V-sector specific-form sub-cases remain ADMISSIBLE-NOT-FORCED. One new CANDIDATE (vacuum-mediated additional CP phases beyond the Jarlskog count for ≥ 3 generations) is opened.

No Phase-2 verdict is overturned. Three Phase-2 closures are structurally enriched. Phase-3 (ED → GR coupling) is substantively unblocked with three new structural input channels: Λ as V1-kernel integral, curvature-vacuum coupling CANDIDATE, and empirical signatures via dispersion-relation modifications and cosmological correlations.

Theorem N1 is ED's first memory-kernel-level structural theorem and the eighth FORCED structural theorem in the ED inventory. It completes the quantum-sector picture at the kernel level alongside Phase-2's form-level theorems while preserving the *form-FORCED, value-INHERITED* methodology.

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